XIAMEN UNIVERSITY MALAYSIA

FINAL EXAMINATION

Course Code:	COM105			
Course Name:	Engineering Physics (I)			
Question Paper Setter:	Dr. Ng Zi Neng			
Academic Session:	2022/04	Question Paper:	A ☑	B ☐
Total No. of Pages:	6	Time Allocated:	2 hours	
Additional Materials:	Appendix			
Apparatus Allowed:	Scientific calculator			

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 6 questions:

Question 1:	5	Sub-questions	Question 4:	3	Sub-questions
Question 2:	3	Sub-questions	Question 5:	4	Sub-questions
Question 3:	3	Sub-questions	Question 6:	3	Sub-questions

   Answer ALL questions.

2. Read the above information carefully to ensure you have the correct and complete question paper.
3. Please follow the requirements of each question and upload all the handwritten corresponding answers on the downloadable answer book provided.
4. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.
5. Candidates are required to compile the answer book in PDF and upload it in Moodle Platform within 15 min after the completion of examination.

(Student ID: _____ Full Name: _____)

QUESTION 1 [CLO1-PLO1:C2, 15 marks]

Explain the followings:

a. If the component of vector G along the direction of vector F is zero, explain and illustrate the two vectors. (3 marks)

b. Rock A is dropped at the same instant that Rock B, is thrown horizontally at the same initial elevation as Rock A. Which will have the greater speed when it reaches ground level? Explain. (3 marks)

c. Explain the force that causes the following: when a driver drives around a corner at high speed, he is pushed toward the side of the car. (3 marks)

d. Explain why is that an axis passing through the centre of mass of an object must be the axis with the smallest moment of inertia. (3 marks)

e. A wood rod and a metal rod each has a piece of paper is wrapped around them. When both of them are held over a flame, the paper in contact with the wood burns first, why? Explain. (3 marks)

QUESTION 2 [CLO3-PLO3:C3, 15 marks]

A wooden crate with the mass of 7.5 kg rests on a plane inclined at 31.0° to the horizontal. The crate is pushed upward on the plane by a force F applied to the object at 43.0° to the horizontal. The coefficients of static and kinetic friction between the block and the plane are, 0.368 and 0.162, respectively. Draw the free body diagram for all the problems below.

a. Solve for the minimum value of F that will prevent the block from slipping down the plane. (5 marks)
b. Solve for the minimum value of F that will start moving the block up the plane? (5 marks)
c. Compute the value of F that will move the block up the plane with constant velocity? (5 marks)

QUESTION 3 [CLO3-PLO3:C3, 15 marks]

Given that the roller coaster cart in Figure 1 has a mass of 523 kg when fully loaded with passengers.

a. If the cart has a speed of 24.0 m/s at Point A, compute for the force exerted by the track on the cart at this point. (5 marks)

b. Assume there is no safety mechanism to keep the cart from flying off the track, solve for the maximum speed the cart can have at Point B and still remain on the track. (5 marks)

c. Solve for the minimum clamping force required for the cart to remain on the track if the cart speed at Point B is 2*x*.0 m/s, where x is the last digit of your ID number (*example: EEE2109388, then v = 28.0 m/s*). What is a negative normal force? (5 marks)

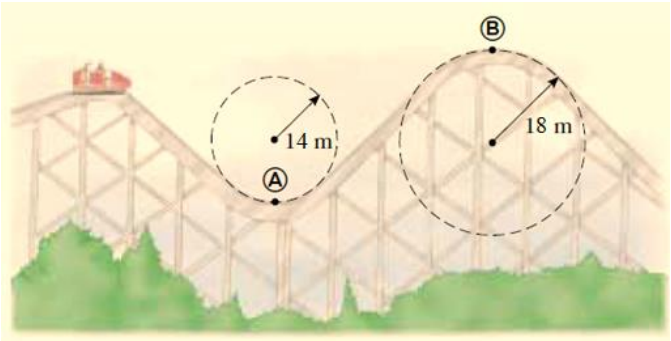


Figure 1

QUESTION 4 [CLO3-PLO3:C3, 20 marks]

a. A plastic pistol uses a spring to project a 6.70-g soft rubber bullet. The spring has a force constant of 9.10 N/m and it is compressed by 5.20 cm before firing. When the gun is fired, the ball moves 13.0 cm through the horizontal barrel of the pistol, and there is a constant friction force of 0.041 N between the barrel and the rubber bullet.

   i. Compute the speed of the rubber bullet as it leaves the barrel of the pistol. (6 marks)

   ii. Compute the maximum speed of the rubber bullet and the point at which it has the maximum speed. (8 marks)

b. A 1.30-m wide and 2.00-m high door has a total mass of 38.0 kg and is initially left open and at rest. Playful Peter threw a 0.450 kg of adhesive gel perpendicular towards the door at 12.0 m/s just before the impact at the centre of the door. Compute the final angular speed of the door. (6 marks)

QUESTION 5 [CLO2-PLO2:C4, 15 marks]

3.20 mol of ideal gas is stored in a cylinder with an adjustable diaphragm at 1.00 atm and 24.0 °C. 15.2 kJ of heat is first added to the gas at constant volume. As the heat continues, the diaphragm moves to allow the gas to expand at constant pressure to twice its original volume. Calculate:

a. the final temperature of the gas. (5 marks)
b. the amount of work done by the gas. (4 marks)
c. the amount of heat added to the gas while it was expanding. (3 marks)
d. the change in internal energy of the gas for the whole process. (3 marks)

QUESTION 6 [CLO3-PLO3:C3, 20 marks]

a. Turbomolecular pumps (TMPs) are kinetic vacuum pumps which operate using a very fast spinning rotor (up to 90,000 RPM). With TMPs it is possible attain pressures of the order of 10^{-14} atm in the laboratory. Consider a volume of air and treat the air as an ideal gas:

   i. Compute the number of molecules that are present in a volume of 1.60 cm^3, a pressure of 5.80×10^{-14} atm, and a temperature of 26°C. (6 marks)

   ii. Compute the number of molecules that would be present at the same temperature but at 1.*x*5 atm instead? Where *x* is the last digit of your ID number (*example: EEE2109388, then p = 1.85 atm*). (7 marks)

b. A Carnot heat engine has a hot reservoir consisting of a large amount of boiling water and a cold reservoir consisting of a large tub of ice and water. In 6 minutes of operation, the heat rejected by the engine melts 80 g of ice. Compute the amount of work W performed by the engine. The heat of fusion for water at 0 °C is approximately 334 kJ per kg. (7 marks)

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